NISHANT VARSHNEY

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EDUCATION

Delhi Technological University (DTU) 2025 - 2029
B.Tech in Computer Science & Engineering **CGPA: 8.95/10**

Kamal Model Sr. Sec. School 2025
Class XII (CBSE) Percentage: 94.4%

Kamal Model Sr. Sec. School 2023
Class X (CBSE) Percentage: 96.8%

EXPERIENCE

Rusborn Aug 2026 - Present
AI & Data Science Intern Delhi, India

- Architected enterprise voice AI platform utilizing **LiveKit Agents SDK, Google Gemma, Deepgram, and Cartesia**, powering natural real-time calls with sub-second voice latency.
- Developed automated campaign orchestration and scheduling in **FastAPI & PostgreSQL**, managing atomic SIP call dispatching, retry logic, and zero-collision appointment booking.
- Delivered a full-stack operations portal using **React 19 & TypeScript**, featuring live call analytics, automated LLM post-call summarization, and dynamic knowledge base management.

PROJECTS

Distill - Autonomous AI Publishing Agent | *Python, LangGraph, FastAPI, Docker, LangSmith* [GitHub]

- Architected an autonomous multi-agent content platform with **LangGraph**, integrating research discovery, deterministic triage, reflection, and QA/revision loops that run continuously in the background.
- Containerized and deployed the **FastAPI** backend on Render, integrating **LangSmith** observability, asynchronous scheduling, **SQLite** + **ChromaDB** hybrid memory, and BM25 retrieval.

Adhikar - Offline AI Legal Assistant | *Python, Flask, Gemma, Ollama, ChromaDB, RAG* [GitHub]

- Built a fully offline, multilingual legal assistant for 11 Indian languages, running Google's **Gemma** locally via Ollama with zero external cloud dependencies.
- Grounded answers via RAG over **6,845 verified chunks** in ChromaDB with claim-by-claim statutory verification, enabling citation-backed responses and zero hallucinations; awarded **Top 4 in Google Gemma Hackathon**.
- Added on-device speech (Faster-Whisper, MMS-TTS) and document OCR (**PaddleOCR**) for automated multimodal evidence extraction.

AIRIS - Hyperlocal Environmental Intelligence | *Python, PyTorch, Graph Transformers, RAG* [GitHub]

- Built a **Graph-Transformer forecasting model** predicting metrics at +24/+48/+72h for 289 Delhi wards by fusing regional pollutant data with ward-level geography topology.
- Achieved **+24% skill** over the persistence baseline using a 4-member ensemble with **conformal prediction intervals** for calibrated uncertainty quantification; integrated a RAG LLM advisor in FastAPI.

Andrew Ng Digital Twin | *Python, Gemini 2.5 Flash, RAG, SQLite Memory, Streamlit* [GitHub]

- Replicated Andrew Ng's pedagogical style using **Gemini 2.5 Flash, RAG, vector embeddings, and SQLite memory**, integrating 5+ AI components for grounded, multi-turn reasoning.
- Implemented curated corpora, browser voice narration, and an evaluation pipeline measuring 5 key quality metrics in an interactive Streamlit UI.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, TypeScript, SQL, HTML, CSS

Machine Learning & DL: PyTorch, Scikit-learn, NumPy, Pandas, CNNs, RNNs, Transformers, Model Training, Evaluation Metrics

Generative & Voice AI: LiveKit Agents SDK, Gemma, Deepgram, Cartesia, RAG, LangGraph, LangChain, ChromaDB, FAISS, Prompt Engineering

Tools & Databases: FastAPI, Flask, PostgreSQL, Supabase, SQLite, Docker, Git, Linux, Streamlit, VS Code

Certifications: Machine Learning Specialization, Deep Learning Specialization (Coursera)

ACHIEVEMENTS & EXTRACURRICULAR

Top 4, Google Gemma Hackathon: Built Adhikar, an offline legal assistant with strict statutory citation verification.

Winner, Green Tag Case Competition, BITS Pilani 2026: Programmed an AI platform predicting real-time disaster risks.

JEE Mains 2025: 99.2 percentile (~14 lakh candidates); qualified JEE Advanced in top percentile.

Academic Excellence: Achieved first semester CGPA of 9.20 at Delhi Technological University.

AIMS-DTU: Member of Artificial Intelligence and Machine Learning Society, DTU.